



TATA

TATA POWER

The Strategic Mandate: Optimizing Infrastructure Investment



Combine operational data, customer usage, and financial ROI metrics to guide investment decisions, optimize station performance, and identify high-potential locations for future expansion.



Performance

Assessing station utilization and energy consumption trends.



Profitability

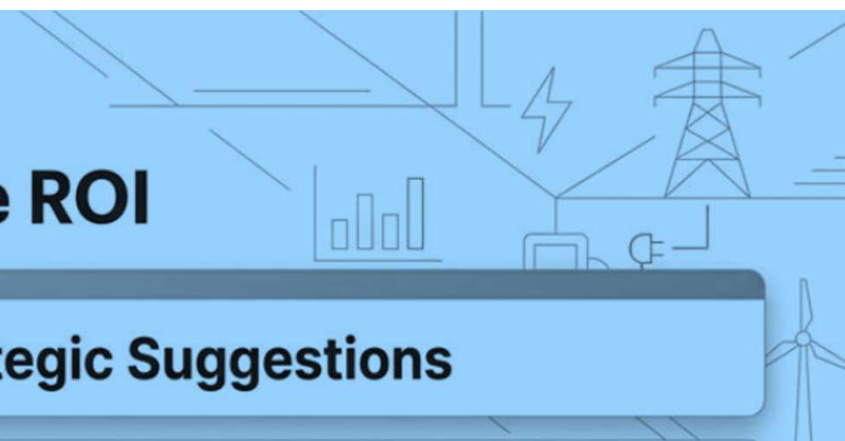
Evaluating CapEx payback periods and EBITDA margins.



Expansion

Identifying high-ROI districts and optimal charger site types (Residential vs. Retail).

Executive Synthesis: Addressing Utilization Imbalances to Maximize ROI



Diagnosed Issues

Uneven Utilization

Massive structural deficits in Jayanagar and Rajajinagar where demand vastly outpaces projected plugs.

CapEx Lags

Mall and Retail sector installations suffer the longest payback periods across the portfolio (5.18 years).

Demand Volatility

Sharp drops in overall energy consumption consistently observed on Fridays and Saturdays.

Strategic Suggestions

Reallocate Resources

Immediately reinforce infrastructure in Jayanagar and Rajajinagar to capture lost revenue.

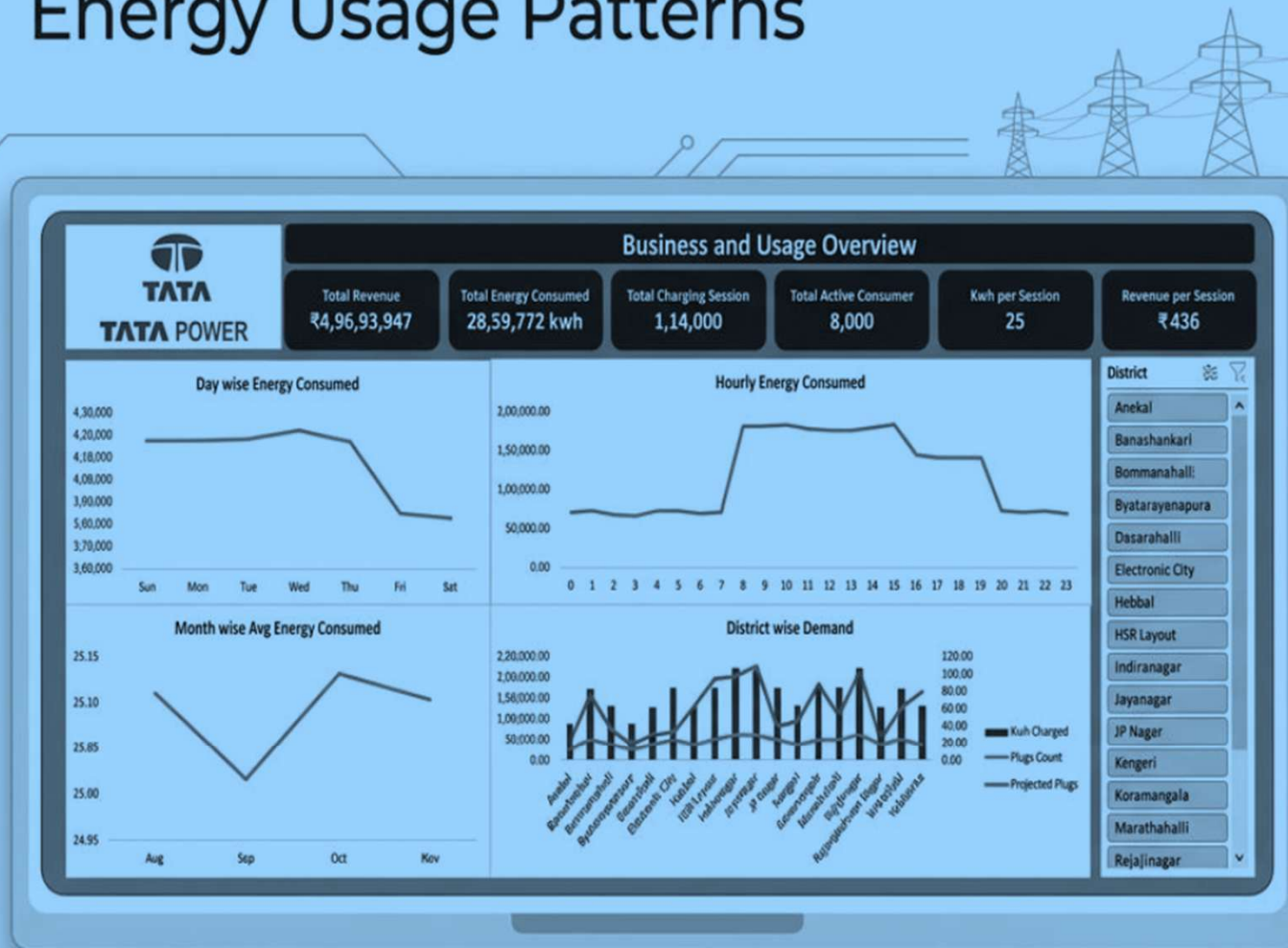
Shift Expansion Strategy

Pivot future CapEx away from Malls and heavily toward Residential Complexes (which boast a 30% faster payback of 3.58 years) and High-ROI districts like Bommanahalli.

Dynamic Pricing

Implement time-of-use pricing models to incentivize charging during low-load Friday/Saturday periods.

Macro Network Performance and Energy Usage Patterns



Analyst Insights

₹4.96 Cr
Total Revenue
across 1.14L sessions
(Avg: ₹436/session)

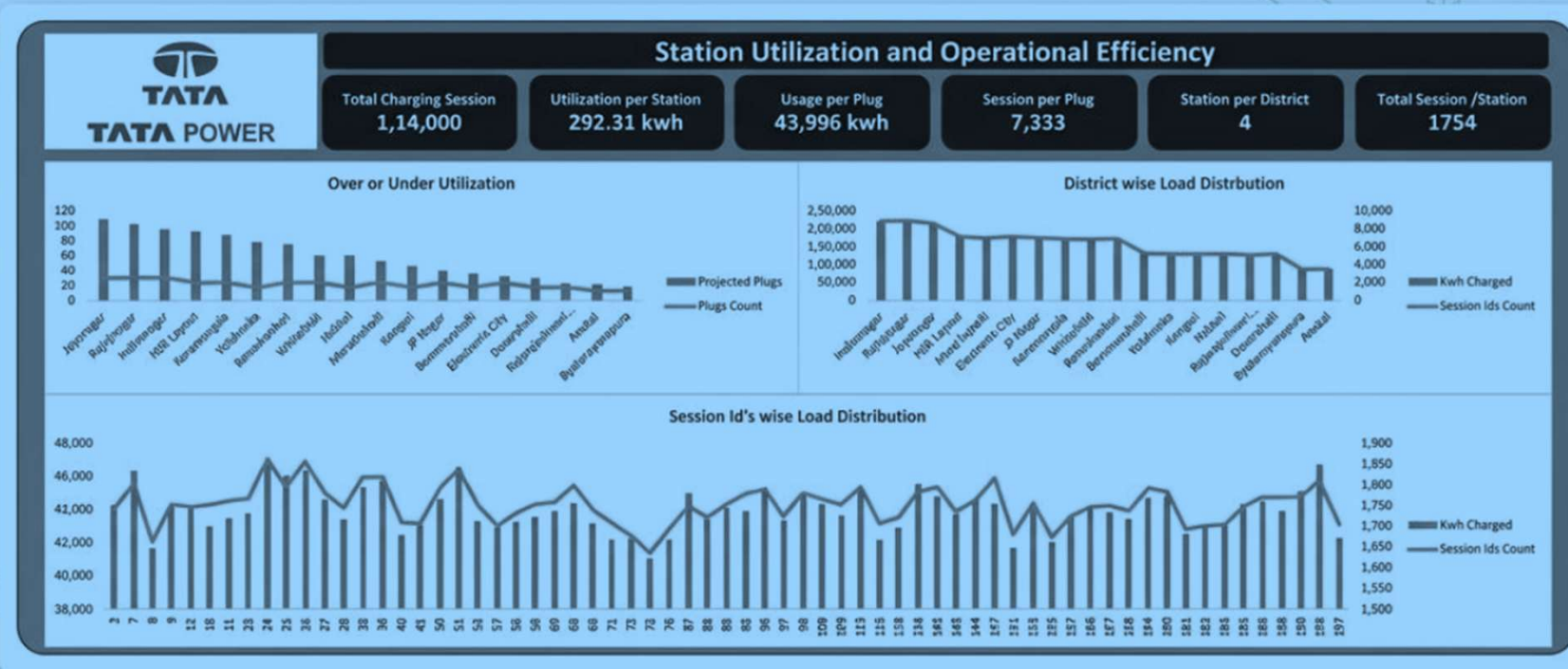
Demand Centers

Jayanagar and Rajajinagar are extreme outliers in district-wise demand, handling exponentially higher kWh loads than neighboring zones.

Temporal Drops

The 'Day-wise' trendline reveals a significant operational gap: sharp dips in network usage on Fridays and Saturdays, requiring weekend demand-generation tactics.

Operational Efficiency: Station Utilization and Plug Deficits



System Averages

- Average utilization: 292.31 kWh / station
- Session volume: 7,333 sessions / plug

Critical Bottleneck Alert

The 'Over or Under Utilization' chart reveals a severe infrastructure deficit in Jayanagar and Rajajinagar. In these districts, the actual charging volume massively eclipses the projected plug count, causing extreme wear and likely customer wait times. Immediate plug reallocation is required here.

Customer Demographics and High-Capacity Demand Drivers

Premium Vehicle Grid Load

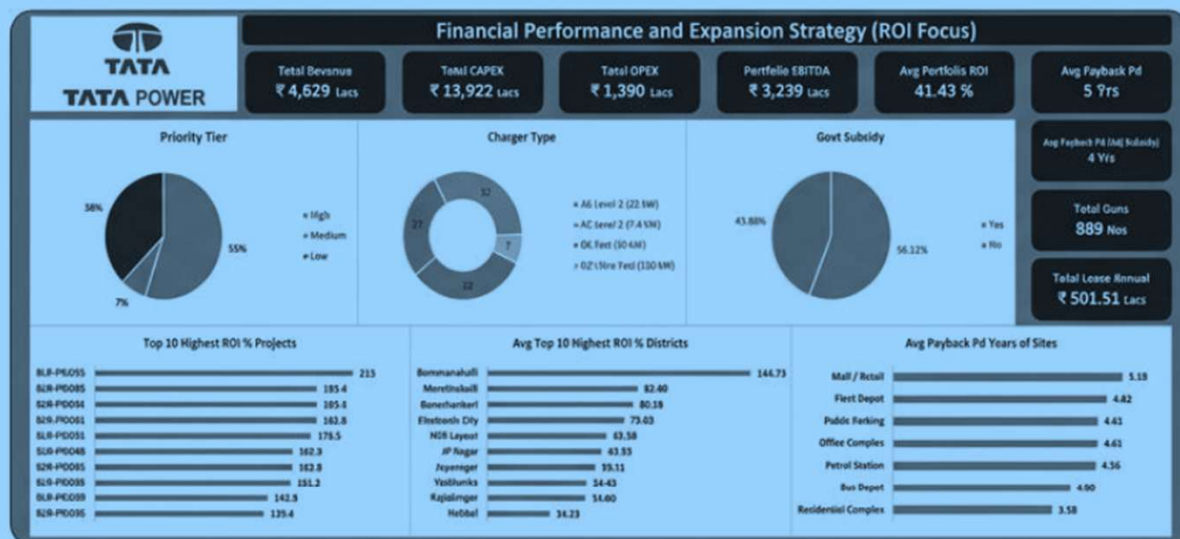
Despite broad demographics, premium capacity vehicles (Audi e-tron 55, Kia EV6) dominate the overall kWh charged metrics, indicating DC fast-charging remains a primary revenue driver.



Mass-Market Adoption Proved

Revenue generation is perfectly balanced across income brackets (35.93% Mid, 32.99% High, 31.08% Low), proving viability across all socio-economic segments.

Financial Returns, Payback Metrics, and CapEx Expansion Tactics



CapEx Expansion Directives

Portfolio Health

Total portfolio ROI sits at a highly healthy 41.43% with EBITDA at ₹3,239 Lacs. Government subsidies successfully shave a full year off the average payback period (5 to 4 years).

District Targeting

Expansion CapEx must immediately route to Bommanahalli, which yields an unmatched 144.73% district ROI.

Site Selection Rule

Halt rapid expansion in Mall/Retail zones (worst payback at 5.18 years). Aggressively pivot to Residential Complex installations (industry-leading payback of 3.58 years).



Thank You

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